



Subject Editor Pieter Rombouts spotlight on analogue-to-digital conversion

toward 'digital' analogue-to-digital converters^{*}

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Analogue-to-digital (A/D) conversion is ubiquitous in the globally interconnected world we live in today. Information travelling over wires, such as electrical signals propagating through the transatlantic telecommunications cables which connect different continents, is inherently analogue. Information travelling wirelessly, such as electromagnetic waves propagating through the air between antennas, has the same property. Since this analogue information needs to be digitally processed, it is apparent that A/D conversion has become one of the cornerstones of modern communication channels in this digital age. Moreover, applications of A/D converters are not limited to communication channels, as they can be found in countless other systems, including sensors, medical imaging and even high-definition video processing. Due to the ever-increasing appetite for more bandwidth and higher signal quality, which is embodied by the push for 5G and new wireline standards, innovation in this area is of paramount importance.

An important enabler for this innovation is technological development by semiconductor manufacturers, who produce progressively smaller and faster devices to keep up with Moore's law. However, with continuously decreasing feature sizes conventional building blocks for A/D converters, such as operational amplifiers, transconductors and comparators, have become increasingly challenging to design, as voltage headroom becomes a scarce resource. This issue triggered an evolution from voltage- to time-domain based conversion techniques and beckoned a more 'digital-friendly' design, which can be easily ported to the newest technology nodes. These considerations seamlessly bring us to voltage-controlled oscillator (VCO)-based A/D conversion.

What is VCO A/D conversion?

Let's start with A/D conversion. Analogue-to-digital conversion is the process where analogue information, or a continuous-time analogue voltage is converted to a digital information stream. This digital signal has a limited amount of values over a certain time frame, or in other words, represents data captured on discrete time instants. In addition, this signal has a limited amount of possible values, which can be represented by a number of bits. Capturing a certain amount of analogue values within a time frame is called 'sampling'. The conversion of this captured analogue value to one of these possible digital values on a certain point in time is called 'quantisation'. Generally, the faster samples are taken, the faster the signal that can be reconstructed, increasing the bandwidth of the system. On the other hand, the more accurate the quantisation, the more accurately the original signal can be reconstructed. Next is the VCO, which is a system which generates an electrical waveform with a frequency proportional to the applied

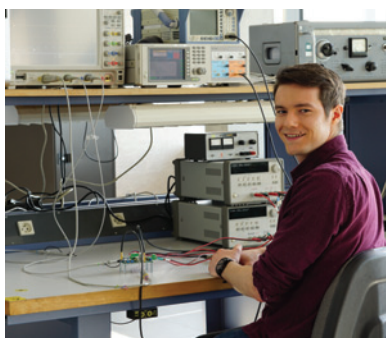


Figure 1: Analogue signals are all around us.

Figure 2: Corresponding author Jonas Borgmans working in the lab at the University of Ghent.

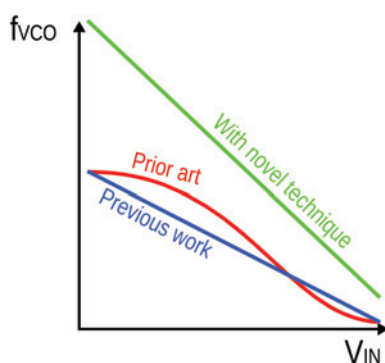
Figure 3: Evolution in voltage-to-frequency conversion curve.

number of cycles or rising edges in one nanosecond, allows a conversion of the analogue voltage to a digital representation, which gives us a VCO-based A/D conversion.

VCOs can be implemented with a ring-oscillator, which is a ring constructed with delay elements that can be as simple as standard digital inverters. This is why it can be interpreted as a 'digital-friendly' circuit. In addition, it shows that it can benefit from technology scaling, as increasingly finer time resolutions can be achieved with faster inverters, improving the signal quality and bandwidth of the system.

Design challenges

An important limitation of a VCO-based A/D conversion is the linearity of the voltage-to-frequency conversion curve. A non-linear conversion curve causes errors to arise when a signal is sent through the system, distorting the signal and thus deteriorating the signal quality and the overall performance of the system. As such, this is an issue that has to be dealt with on circuit or architectural level. Another demanding challenge is achieving continuously higher oscillation frequency to keep up with bandwidth demands. Reaching these higher oscillation speeds requires faster devices and smarter designs. In addition, obtaining a high linearity gets increasingly difficult with growing oscillation frequencies. Figure 2 shows the evolution in voltage-to-frequency conversion curves. The red curve, representing prior art VCOs, is visibly curved when ideally it should be a straight line. This is why typically only a small section of the conversion curve was used to obtain an approximately linear system. In previous work of this research team, an extremely linear VCO was developed, which is depicted by the blue



voltage at the input. Using this VCO, information can be converted from the voltage-domain to the temporal-domain, as the input information is encoded in the phase of the output waveform. Quantising the frequency or period of the generated waveform, for instance by counting the

straight line. Finally, the green line shows what the research team has achieved, maintaining a very high linearity while increasing the oscillation frequency substantially.

Research achievements

In the featured article, a new technique is proposed to improve the bandwidth and noise performance of extremely linear voltage-controlled ring oscillators. The technique is a result from a co-optimization of noise and linearity performance, in contrast to other works, which usually only focus on one of these aspects. The key element is the modified delay cell, where the traditional cross-coupled inverters are modified into 'feed-forward' coupling inverters that pre-charge the subsequent elements in the ring. This pre-charging mechanism generates higher oscillation frequencies while retaining an excellent linearity and noise performance, resulting in a greatly enhanced performance. Another advantage of this technique is that the circuitry is essentially an interconnection of

inverters, allowing the design to be scaled when faster technologies are available, providing even higher-speed designs in the future.

The defining concept of this article however, is that using this novel circuit-level technique removes the trade-off between performance and robustness of voltage-controlled ring oscillators. This realization is what sets this letter apart from previous works on the same subject.

In conclusion, the research team believes that applying this technique sets a new standard for high-speed VCO structures that require a high linearity and a low power consumption, due to its simple and 'digital-friendly' design and excellent properties.

More to come

The improved VCO performance realized by this new circuit-level technique enables extremely linear open-loop VCO-ADC systems, even with very high oscillation frequencies. As such, this work could be a catalyst for more research on high-speed VCO-based ADCs, accommodating the continuously increasing demand for higher

bandwidths. In addition, as a 'digital-friendly' design is obtained, the proposed VCO structure can be used for years to come, as it can easily be scaled to newer and faster technology nodes. Furthermore, VCOs using this new technique can be applied in other areas, such as phase-locked loops (PLLs) commonly used in FM applications.

Following the trend of more time-based A/D conversion techniques, the featured authors have already started to embed this VCO in a VCO-ADC architecture, aiming toward 500MHz bandwidth designs with a low power consumption and excellent linearity. This course of action is especially important considering the newest CMOS technologies (e.g. 14 nm technology node), where the switching speed is even further increased. However, this increase in speed is at the price of sacrificing design freedom concerning the layout, which further pushes the need for all-digital circuits. This work anticipates this trend, by enabling simple, digital-friendly A/D converters which can maximally profit from the advantages of these new technologies.